

Thesis Defense:
Invariant measures for the stochastic heat equation with
no drift term

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The stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x), & t \geq 0, x \in \mathbb{R}^d \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The Stochastic Heat Equation is interpreted as follows:

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y)b(u(s, y))W(ds, dy)$$

where $G(t, x)$ is the heat kernel

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right)$$

- A processes, $\{u(t, x) : t \geq 0, x \in \mathbb{R}^d\}$, that satisfies this relation is referred to as the *mild solution*.

The noise term $\dot{W}$

- $W = \{W(\phi) : \phi \in \mathcal{S}(\mathbb{R}^d)\}$ is a centered Gaussian process uniquely defined through the following covariance functional:

$$J(\phi, \psi) := \mathbb{E}[W(\phi)W(\psi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x),$$

where $\tilde{\psi}(s, x) = \psi(s, -x)$. We have the following nomenclature.

- *Space-time white noise*: $\Gamma(dx) = \delta_0(dx)$.
- *Time white and spatially colored*: $\Gamma(dx) = f(x)dx$ where f is a non-negative and non-negative definite function referred to as the *Correlation function* and its Fourier transform,

$$\hat{f}(\xi) := \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-ix \cdot \xi) f(x)dx,$$

is referred to as the *spectral density*.

Existence and uniqueness with space-time white noise

- A necessary condition for global existence, *Dalang's condition* [1], is the following:

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) < \infty, \quad \forall t > 0.$$

- Direct calculations yields that,

$$\int_0^t ds \int_{\mathbb{R}^d} dy G^2(s, y) = \frac{\Gamma(d/2)}{2(2\pi)^d} \int_0^t \frac{1}{s^{d/2}} ds,$$

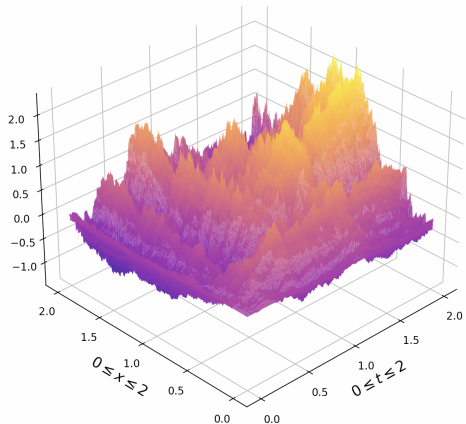
which forces,

$$d/2 < 1 \quad \Longleftrightarrow \quad d < 2.$$

- Thus a global space-time white noise solution may only exist when the spatial dimension is 1.

Proposition (DaPrato and Zabczyk [2, Propsition 4.19])

A space-time white noise, dB , is the distributional derivative of a Brownian sheet, $B(t, x)$.



White noise expansion DaPrato [2]

If $t \geq 0$ and $0 \leq x \leq \pi$ then,

$$B(t, x) = \sum_{n=0}^{\infty} \frac{\beta_n(t)}{n + \frac{1}{2}} \sqrt{\frac{2}{\pi}} \sin \left[\left(n + \frac{1}{2} \right) x \right]$$

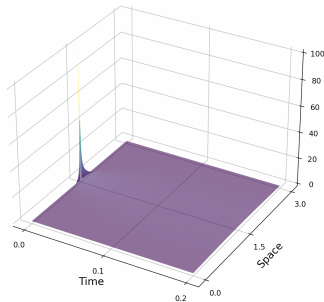
Walsh [4]

$$\int_t^{t+\delta_t} \int_x^{x+\delta_x} dB(ds, dy) \sim N(0, \delta_t \delta_x)$$

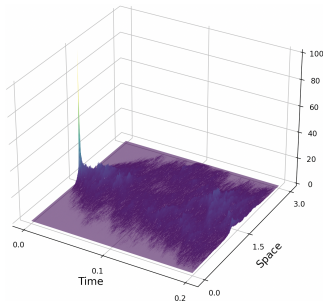
Brownian Sheet Simulation (Walsh [4]):

$$B(t, x) \sim B(t - \delta_t, x) + B(t, x - \delta_x) - B(t - \delta_t, x - \delta_x) + N(0, \delta_t \delta_x)$$
$$B(t, 0) = B(0, x) = 0 \text{ for all } t, x > 0.$$

$$u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W} \text{ with } u_0(x) = \delta_0(x) \quad (\text{white noise})$$



$\lambda = 0$

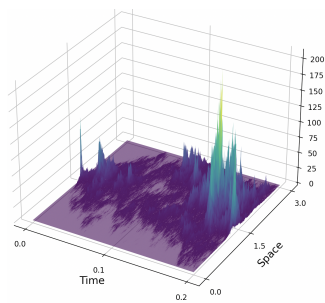
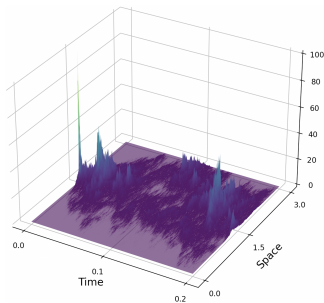


$\lambda = 1$

- Simulations done by using a finite difference scheme [5].
- We partition (space) $[0, 3]$ into n_x pieces of size $\delta_x = 3/n_x$.
- We partition (time) $[0, .2]$ into n_t pieces of size $\delta_t = .2/n_t$.
- To control error, we assume $n_x^2 \delta_t < \frac{1}{2}$ which implies $n_t \gg n_x$.

$$u(t + \delta_t, x) = u(t, x) + n_x^2 \delta_t (u(t, x + \delta_x) + u(t, x - \delta_x) - 2u(t, x)) \\ + n_x \lambda u(t, x) \text{normal}(0, \delta_t \delta_x).$$

Intermittency of $u_t - \frac{1}{2}u_{xx} = \lambda u \dot{W}$ (white noise)



● Notice the enlargement of the peaks as λ increases. We have the following theorem.

Theorem (Chen and Dalang [1, Theorem 2.4])

The second moments of the stochastic heat equation driven by space-time white noise grow exponentially in time.

Will the second moments ever be bounded in time?

- Several slides below will require the second moments to be bounded in time:

$$\sup_{t \geq 0} \|u(t, x)\|_2 < \infty, \quad \text{where } \|X\|_p = [\mathbb{E}(|X|^p)]^{1/p}.$$

- This is possible, however this will require us to only consider $d \geq 3$, and therefore we must only consider a spatially colored noise.
- Recall from above, that the covariance of a spatially colored noise is given by the following:

$$\begin{aligned} \mathbb{E}[W(\phi)W(\psi)] &= \int_0^\infty ds \int_{\mathbb{R}^d} dx f(x)(\phi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^{2d}} dx dy \phi(s, x) f(x - y) \psi(s, y) \\ &= (2\pi)^{-d} \int_0^\infty ds \int_{\mathbb{R}^d} d\xi \hat{f}(\xi) \mathcal{F}\phi(s, \xi) \overline{\mathcal{F}\psi(s, \xi)} \end{aligned}$$

$$u_t - \frac{1}{2} u_{xx} = b(u) \dot{W} \text{ with } u(0, \cdot) = \mu(\cdot) \quad (\text{colored noise})$$

Theorem (Chen and Huang [2, Theorem 1.2])

There exists a unique global mild solution under the following:

- $b : \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a $L_b, L_0 > 0$ so for all $u, v \in \mathbb{R}$ and $x \in \mathbb{R}^d$,

$$|b(x, u) - b(x, v)| \leq L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0.$$

- The initial measure, $\mu(dx)$ (resp. $\varphi(x)dx$), satisfies for some $a > 0$,

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty \quad \left(\text{resp. } \int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty \right),$$

where $|\mu| = \mu^+ + \mu^-$ is the *Hahn decomposition*.

- The spectral density, $\hat{f}$, satisfies *Dalang's condition* for some $\beta > 0$:

$$\Upsilon(\beta) := \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty.$$

Bounded moments in time

Theorem (Chen and Kim [4, Theorem 1.3 and Lemma 2.5])

Suppose that we strengthen Dalang's condition and the Lipschitz assumption to the following:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Then the following holds:

$$\begin{aligned} \|u(t, x)\|_2 &\leq C_2 \left[1 + \left(G(t, \cdot) * |\mu| \right)(x) \right] \\ &= C_2 \left[1 + \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right], \quad \text{for all } t \geq 0. \end{aligned}$$

- If $\mu = \delta_0$ then $\|u(t, x)\|_2 \leq C_2 [1 + (2\pi t)^{-d/2}]$.
- Later, it will suffice consider μ such that $\sup_{t \geq t_0} \|u(t, x)\|_2 < \infty$.

Transition functions and their semigroup

Definition (Markovian Transition Functions)

- Let H be a Hilbert space and $\mathcal{B}(H)$ be the borel sets of H .
- $\{P_t(x, \Gamma) : t \geq 0, x \in H, \Gamma \in \mathcal{B}(H)\}$ forms a family of *Markovian transition functions* if,

(1) $x \mapsto P_t(x, \Gamma)$ if measurable,

(2) $\Gamma \mapsto P_t(x, \Gamma)$ is a probability measure on H ,

(3) $P_{t+s}(x, \Gamma) = \int_H P_t(x, dy) P_s(y, \Gamma)$,

(4) $P_0(x, \Gamma) = \mathbf{1}_\Gamma(x)$.

Definition (Transition Semigroup)

The above transition functions form the following semi group on $\mathcal{B}(H)$, which denotes the bounded Borel measurable functions on H :

$$P_t \varphi(x) = \int_H \varphi(y) P_t(x, dy), \quad \varphi \in \mathcal{B}(H).$$

Invariant measures

Definition (Invariant measures)

- A probability measure, η , on H is said to be *invariant* for the transition functions P_t if for all $\Gamma \in \mathcal{B}(H)$ and $t > 0$,

$$\int_H P_t(x, \Gamma) \eta(dx) = \int_H P_0(x, \Gamma) \eta(dx) = \int_H \mathbf{1}_\Gamma(x) \eta(dx) = \eta(\Gamma),$$

which is equivalent to,

$$\int_H P_t \varphi(x) \eta(dx) = \int_H \varphi(x) \eta(dx), \quad \text{for all } \varphi \in \mathcal{B}(H).$$

- This implies that the η -average of $x \mapsto P_t(x, \Gamma)$ is $\eta(\Gamma)$ and it is independent of time:

$$\text{AVG}_\eta[P_t(\cdot, \Gamma)] = \frac{1}{\eta(H)} \int_H P_t(x, \Gamma) \eta(dx) = \int_H P_t(x, \Gamma) \eta(dx) = \eta(\Gamma).$$

Invariant measures for the stochastic heat equation

- First we define the law of the solution as follows:

$$\mathcal{L}(u(t, \cdot; \zeta))(\Gamma) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \zeta) \in \Gamma], \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H),$$

and for the mean time we are intentionally vague about H .

- Our intention is for

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma)$$

to form a Markovian semigroup.

- Established theory by DaPrato and Zabczyk [2] requires the solution,

$$\zeta \mapsto u(t, \cdot; \zeta),$$

to be almost surely H -valued for $t > 0$ and $\zeta \in H$.

- So what is H ?

$H = L^2_\rho(\mathbb{R}^d)$: weighted square integrable functions

Definition (Admissible Weight)

A positive, bounded and continuous $L^1(\mathbb{R}^d)$ function, ρ , is an *admissible weight* if

$$\frac{\left(G(t, \cdot) * \rho(\cdot)\right)(x)}{\rho(x)} \leq C_\rho(T), \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d.$$

Definition ($L^2_\rho(\mathbb{R}^d)$)

The space $H := L^2_\rho(\mathbb{R}^d)$ is the space of ρ -weighted square integrable functions:

$$\|\zeta\|_\rho^2 := \int_{\mathbb{R}^d} |\zeta(x)|^2 \rho(x) dx < \infty.$$

Moreover, H is a Hilbert space with inner product defined as

$$\langle \zeta, \psi \rangle_\rho := \int_{\mathbb{R}^d} \zeta(x) \psi(x) \rho(x) dx.$$

Weight examples

- The following two are examples of admissible weights:

$$\exp(-a|x|), \text{ for } a > 0 \quad \text{and} \quad \frac{1}{1+|x|^a}, \text{ for } a > d.$$

- One can show that the following is not admissible for any $a > 0$,

$$\exp(-a|x|^b), \text{ for all } b > 1.$$

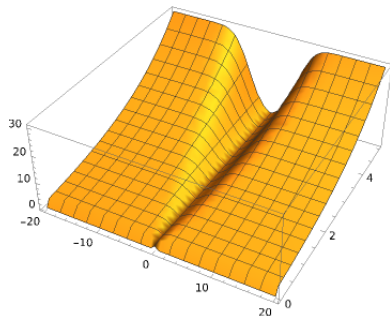


Figure: $x \mapsto \frac{(G(t, \cdot) * \exp(-|\cdot|))(x)}{\exp(-|x|)}$

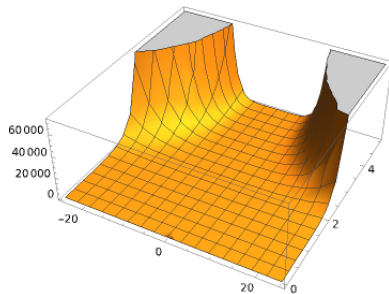


Figure: $x \mapsto \frac{(G(t, \cdot) * \exp(-|\cdot|^{1.25}))(x)}{\exp(-|x|^{1.25})}$

The solution is almost surely H -valued.

Theorem

Let ρ be an admissible weight and suppose that

$$\Upsilon(0) = \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi)}{|\xi|^2} d\xi < \infty \quad \text{and} \quad L_b^2 < \frac{1}{128\Upsilon(0)}.$$

Now suppose our initial measure, μ , satisfies the following:

$$\mathcal{G}_\rho(t; |\mu|) := \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy) \right)^2 \rho(x) dx < \infty.$$

Then for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; 1 + |\mu|) < \infty.$$

In particular, for any $\zeta \in H$ and for all $T > 0$, there exists a C_T such that

$$\mathcal{G}_\rho(t; \zeta) \leq 2 \|\rho\|_{L^1(\mathbb{R}^d)} + 2C_T \|\zeta\|_\rho < \infty, \quad \text{for all } t \in [0, T].$$

Invariant measures for the stochastic heat equation

- So the solution acts as an operator, mapping H to H .
- Moreover, the Markov property of the solution [2, Th 9.21] implies that

$$P_t(\zeta, \Gamma) := \mathcal{L}(u(t, \cdot; \zeta))(\Gamma), \quad \text{for all } \zeta \in H \text{ and } \Gamma \in \mathcal{B}(H)$$

for a family of Markovian transition functions.

Definition

A measure η is *invariant* for the stochastic heat equation if η is invariant for the transition functions P_t . In other words,

$$\int_H P_t(\zeta, \Gamma) \eta(d\zeta) = \int_H \mathcal{L}(u(t, \cdot; \zeta))(\Gamma) \eta(d\zeta) = \eta(\Gamma), \quad \text{for all } \Gamma \in \mathcal{B}(H).$$

Weak convergence of measures

- Let $\mathcal{M}(H)$ and $C_b(H)$ denote the probability measures and bounded continuous functions on H .
- For $\varphi \in C_b(H)$ and $\nu \in \mathcal{M}(H)$,

$$\langle \varphi, \nu \rangle = \int_H \varphi(\zeta) \nu(d\zeta).$$

Definition

We say ν_n converges *weakly* to η in $\mathcal{M}(H)$ if for all $\varphi \in C_b(H)$,

$$\langle \varphi, \nu_n \rangle \rightarrow \langle \varphi, \nu \rangle, \quad \text{as } n \rightarrow \infty.$$

We will use the notation, $\nu_n \rightarrow \nu$, to denote the weak convergence of measures.

Some existence results

Theorem (DaPrato [2, Propositions 11.2 and 11.3])

If $\zeta_1, \zeta_2 \in H$ and $\eta_1, \eta_2 \in \mathcal{M}(H)$ are such that

$$\mathcal{L}\left(u(t, \cdot; \zeta_1)\right) \rightarrow \eta_1 \text{ as } t \rightarrow \infty \quad \text{or} \quad \frac{1}{t_n} \int_0^{t_n} \mathcal{L}\left(u(s, \cdot; \zeta_2)\right) ds \rightarrow \eta_2 \text{ as } n \rightarrow \infty,$$

for some $t_n \uparrow \infty$, then both η_1 and η_2 are invariant.

- Red implies Blue but not vice versa.
- For example, suppose $\eta_t(A) \rightarrow \eta(A)$ as $t \rightarrow \infty$. Then for $\epsilon > 0$, there exists N so for all $t > N$ we have the following:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n \eta_t(A) dt \leq \lim_{n \rightarrow \infty} \frac{1}{n} \left(\int_0^N \eta_t(A) dt + \int_N^n \eta(A) \pm \epsilon dt \right),$$

thus,

$$\eta(A) - \epsilon \leq \lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n \eta_t(A) dt \leq \eta(A) + \epsilon.$$

Under what setup has an invariant measure been proven to exist?

- Research on invariant measures for $u_t - \frac{1}{2}\Delta u = f(u) + b(u)\dot{W}$ on the whole space with a globally Lipschitz b is not vast.
- Tessitore and Zabczyk [3] is the only other paper which studies $u_t - \frac{1}{2}\Delta u = b(u)\dot{W}$ on the whole space with with a globally Lipschitz b .
- Tessitore and Zabczyk showed there exists an $\eta \in \mathcal{M}(H)$ such that,

$$\frac{1}{t_n} \int_1^{t_n} \mathcal{L}\left(u(s, \cdot; 1)\right) ds \rightarrow \eta, \quad \text{as } n \rightarrow \infty,$$

under the following assumption on the spectral density $\widehat{f}$:

$$\int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| * \left| \mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \right| \right](\xi) d\xi < \infty.$$

- If one could find a f such that $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ and $\widehat{f} \geq 0$ then

$$\begin{aligned}
 & \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| * \left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| \right] (\xi) \, d\xi \\
 &= \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) * \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right] (\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{2-d} \left[\mathcal{F}^{-1} \left(\widehat{f} \right) \right] (\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{2-d} f(\xi) \, d\xi \\
 &= K \int_{\mathbb{R}^d} |\xi|^{-2} \widehat{f}(\xi) \, d\xi = \Upsilon(0),
 \end{aligned}$$

which is precisely Dalang's condition.

- $\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \geq 0$ is restrictive.

- $\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right) \geq 0$ requires $\sqrt{\widehat{f}}$ to be non-negative definite and this may not be true even if $\widehat{f}$ is non-negative and non-negative definite.

Example

Consider the pair

$$f(x) = (2\pi)^{-1} \sin(x^2)/x^2 \quad \text{and} \quad \widehat{f}(\xi) = \frac{1}{4} \max\{2 - |\xi|, 0\}.$$

Both f and $\widehat{f}$ is clearly non-negative and thus both are non-negative definite. However, direct calculation shows that

$$\mathcal{F}^{-1}\left(\sqrt{\widehat{f}}\right)(x) = \frac{\sqrt{\frac{\pi}{2}} \left(-\cos(2x) \mathcal{S}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) + \sin(2x) \mathcal{C}\left(\frac{2\sqrt{x}}{\sqrt{\pi}}\right) \right)}{x^{3/2}}$$

Where $\mathcal{S}$ and $\mathcal{C}$ are the sine and cosine Fresnel integrals,

$$\mathcal{C}(z) = \int_0^z \cos\left(\frac{\pi t^2}{2}\right) dt \quad \text{and} \quad \mathcal{S}(z) = \int_0^z \sin\left(\frac{\pi t^2}{2}\right) dt.$$

$\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ takes positive and negative values

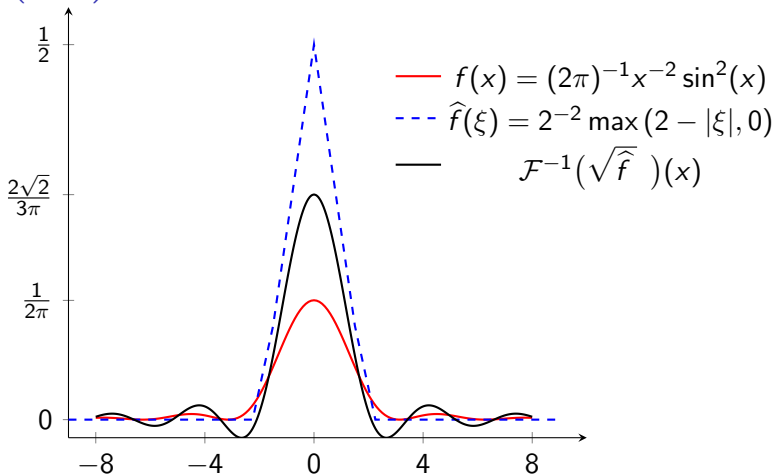


Figure: One example that $\mathcal{F}^{-1}(\sqrt{\widehat{f}})(x)$ assumes both positive and negative values.

● So a simpler condition must be formulated in order to give a full example.

● Recall that

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_{\rho} \right) \leq C \mathcal{G}_{\rho}(t; \mu) := C \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y)(1 + |\mu|)(dy) \right)^2 \rho(x) dx$$

Theorem

Suppose that the initial measure, μ , the diffusion term, b , and the spectral density, $\hat{f}$ satisfy

$$\Upsilon_{\alpha}(0) := \int_{\mathbb{R}^d} \frac{\hat{f}(\xi) d\xi}{|\xi|^{2(1-\alpha)}} < \infty, \quad \limsup_{t>0} \mathcal{G}_{\rho}(t; \mu) < \infty, \quad \text{and} \quad L_b^2 < \frac{1}{2^7 \Upsilon(0)},$$

for some $\alpha \in (2^7 \Upsilon(0) L_b^2, 1)$ and also for $\alpha = 0$. Then there exists an invariant measure for the stochastic heat equation. More precisely, the following weak limit exists for each $t_0 > 0$:

$$\frac{1}{t_0} \int_{t_0}^{t_n+t_0} \mathcal{L} \left(u(s, \cdot; \mu) \right) ds \rightarrow \eta_{t_0, \mu},$$

where η_{t_0} is the invariant measure.

Strategy

- The strategy is to prove tightness of the laws: when given $\epsilon \in (0, 1)$, to prove the existence of a compact set $\mathcal{K}$ such that

$$\mathcal{L}u(t, \cdot; \mu)(\mathcal{K}) \geq 1 - \epsilon \quad \text{for all } t \geq t_0 > 0.$$

- This implies that

$$A_n(\mathcal{K}) := \frac{1}{n} \int_{t_0}^{n+t_0} \mathcal{L}\left(u(s, \cdot; \mu)\right)(\mathcal{K}) ds \geq \frac{1}{n} \int_{t_0}^{n+t_0} 1 - \epsilon ds = 1 - \epsilon,$$

and so the sequence of measures, A_n is tight.

- Every tight sequence of measures has a weakly convergent subsequence that converges to an invariant measure, η . Hence, there exists $(t_n) \uparrow \infty$ such that,

$$A_{t_n} \rightarrow \eta$$

- Unfortunately this does not imply that

$$\mathcal{L}u(t, \cdot; \mu) \rightarrow \eta \quad \text{as } t \rightarrow \infty.$$

$\widehat{f}(\xi)$ and $\Upsilon_\alpha(0)$

- It is easy to verify that

$$\widehat{f}(\xi) = \prod_{i=1}^d \max\{2 - |\xi_i|, 0\} \quad \text{and} \quad f(x) := \prod_{i=1}^d (2\pi)^{-1} \sin(x_i^2)/x_i^2$$

are allowable using our condition above. Indeed, for any $\alpha \in [0, 1)$,

$$\begin{aligned} \Upsilon_\alpha(0) &= C \int_{\mathbb{R}^d} \frac{\prod_{i=1}^d \max\{2 - |\xi_i|, 0\}}{|\xi|^{2(1-\alpha)}} d\xi \\ &\leq C \int_{|\xi| \leq 2\sqrt{d}} \frac{2^d}{|\xi|^{2(1-\alpha)}} d\xi \\ &= C \int_0^{2\sqrt{d}} \frac{r^{d-1}}{r^{2(1-\alpha)}} dr. \end{aligned}$$

- Hence, if $\alpha > 1 - d/2$, then $\Upsilon_\alpha(0) < \infty$

A comment about $\Upsilon_\alpha(0) < \infty$

- For $\alpha \in (0, 1/2)$, $\Upsilon_{2\alpha}(0) < \infty$ implies Hölder continuity of the solution [2].
- This condition may be slightly "weakened" to the following:

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0 \text{ and some } \alpha \in \left(0, \frac{1}{2}\right)$$

Lemma

We have the following relations between $\Upsilon_\alpha(0)$, $\Upsilon(0)$ and $\mathcal{H}_\alpha(t)$:

- (1) *However, $\Upsilon_{2\alpha}(0) < \infty \implies \mathcal{H}_\alpha(t) < \infty$ for all $t > 0$.*
- (2) *$\lim_{t \rightarrow \infty} \mathcal{H}_\alpha(t) = \Gamma(1 - 2\alpha)\Upsilon_{2\alpha}(0)$.*
- (3) *When $\Upsilon(0) < \infty$ then $\Upsilon_{2\alpha}(0) < \infty \iff \mathcal{H}_\alpha(t) < \infty$ for all $t > 0$.*

Permissible spectral density

- Recall that $\widehat{f}$ must satisfy for both $\alpha = 0$ and $\alpha \in (2^7 \Upsilon(0) L_b^2, 1)$ the following:

$$\Upsilon_\alpha(0) := \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi) \, d\xi}{|\xi|^{2(1-\alpha)}} < \infty.$$

- Recall that $d \geq 3$ and so $\widehat{f}$ may have slow blow up at the origin but must have fast decay at $\pm\infty$ to counter act the thick tails of $|\xi|^{2(1-\alpha)}$.
- This excludes the Riesz kernel.
- However, the Bessel kernel has polynomial blowup at the origin and exponential decay at $\pm\infty$ and will be permissible.

Bessel kernel

- Let f_s denote the Bessel kernel. When $0 < s < d$ we have that

$$f_s(x) \sim \exp(-|x|/2) \text{ for } |x| > 2 \quad f_s(x) \sim \frac{1}{|x|^{d-s+2}} \text{ for } |x| < 2.$$

- The Fourier transform $\hat{f}$ is given by $\hat{f}(\xi) = (1 + |\xi|^2)^{-s/2}$.

Example

When considering the Bessel kernel, direct calculation yields that

$$\Upsilon_\alpha(0) = \frac{\Gamma\left(\frac{d}{2} - 1 + \alpha\right) \Gamma\left(\frac{s-d}{2} + 1 - \alpha\right)}{2^d \pi^{d/2} \Gamma(s/2)},$$

and when $\alpha = 0$, this simplifies to

$$\Upsilon(0) = \frac{\Gamma\left(\frac{2+s-d}{2}\right)}{2^{d-1} \pi^{d/2} (d-2) \Gamma(s/2)},$$

and we must assume that

$$0 < \alpha < 1 - \frac{(d-s)_+}{2} \quad \text{and} \quad s > d-2 > 0.$$

Quick verification for the Bessel kernel

- Recall that $\Upsilon_\alpha(0) = C \int_{\mathbb{R}^d} \frac{\widehat{f}(\xi)}{|\xi|^{2(1-\alpha)}} d\xi$ and $\widehat{f}(\xi) = (1 + |\xi|^2)^{-s/2}$.

$$\begin{aligned}\Upsilon_\alpha(0) &= C \int_{\mathbb{R}^d} \frac{1}{|\xi|^{2(1-\alpha)} (1 + |\xi|^2)^{s/2}} d\xi \\ &= C \int_0^\infty \frac{z^{d-1}}{z^{2(1-\alpha)} (1 + z^2)^{s/2}} dz,\end{aligned}$$

- Now by the change of variables $z = r^2/(1 + r^2)$, we can evaluate the above integral explicitly by transforming it to the Beta

$$\begin{aligned}C \int_0^\infty \frac{r^{d-1}}{r^{2(1-\alpha)} (1 + r^2)^{s/2}} dr &= C \int_0^1 z^{d/2+\alpha-2} (1 - z)^{(s-d)/2-\alpha} dz \\ &= C \frac{\Gamma(d/2 - 1 + \alpha) \Gamma((s-d)/2 + 1 - \alpha)}{2\Gamma(s/2)},\end{aligned}$$

Riesz type kernel

- The g_s denote the Riesz kernel for $0 < s < d$: $g_s(x) = |x|^{-s}$.

- g_s will not satisfy

$$\Upsilon(0) = C \int_{\mathbb{R}^d} \widehat{g}_s(\xi) |\xi|^{-2} d\xi = \int_{\mathbb{R}^d} |\xi|^{d-s} |\xi|^{-2} d\xi < \infty,$$

since this requires $d - s - 2 > d$ and $d - s - 2 < d$.

- Instead, we can use the following

$$\widehat{r}_s(\xi) := \widehat{r}_{s_1, s_2}(\xi) = \widehat{f}_{s_1}(\xi) + f_{s_2}(\xi),$$

where $0 < s_i < d$ for $i = 1, 2$ and f_{s_1} and f_{s_2} are Bessel kernels.

- $\widehat{r}_s$ has polynomial growth at the origin and polynomial decay at infinity:

$$\widehat{r}_s(\xi) \sim \begin{cases} |\xi|^{-(d-s_2)} & |\xi| \rightarrow 0 \\ |\xi|^{-s_1} & |\xi| \rightarrow \infty, \end{cases}$$

Riese type kernel

- Notice that

$$\begin{aligned}\gamma_\alpha(0) &= \int_{\mathbb{R}^d} \frac{\widehat{r_s}(\xi)}{|\xi|^{2(1-\alpha)}} d\xi \\ &= \int_{\mathbb{R}^d} \frac{\widehat{f_{s_1}}(\xi)}{|\xi|^{2(1-\alpha)}} d\xi + \int_{\mathbb{R}^d} \frac{f_{s_2}(\xi)}{|\xi|^{2(1-\alpha)}} d\xi \\ &= \int_{\mathbb{R}^d} \frac{\widehat{f_{s_1}}(\xi)}{|\xi|^{2(1-\alpha)}} d\xi + (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f_{s_2}}(\xi)}{|\xi|^{d-2(1-\alpha)}} d\xi\end{aligned}$$

- We can easily calculate that

$$\gamma_\alpha(0) = \frac{\Gamma\left(\frac{d}{2} - 1 + \alpha\right) \Gamma\left(\frac{s_1 - d}{2} + 1 - \alpha\right)}{2^d \pi^{d/2} \Gamma(d/2) \Gamma(s_1/2)} + \frac{\Gamma(1 - \alpha) \Gamma\left(\alpha - 1 + \frac{s_2}{2}\right)}{2^{2d} \pi^{3d/2} \Gamma(d/2) \Gamma(s_2/2)} < \infty,$$

where the finiteness requires us to assume that

$$0 < d - 2(1 - \alpha) < s_1 < d \quad \text{and} \quad 0 < 2(1 - \alpha) < s_2 < d.$$

- It is not clear how to calculate $\mathcal{F}^{-1}\left(\sqrt{\widehat{r}(\cdot)}\right)(x)$.

Allowable Initial Conditions:

(1) For any $\zeta \in L^\infty(\mathbb{R}^d)$ then for all $t \geq 0$

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq \left(1 + \|\zeta\|_{L^\infty(\mathbb{R}^d)} \right)^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

(2) For $\delta_0(\cdot)$ and for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; \delta_0)\|_\rho^2 \right) \leq (2 + G(t, 0)^2) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty.$$

(3) Consider $|\cdot|^{-\lambda}$ for any $\lambda \in (0, d)$. Then for all $t > 0$,

$$\mathbb{E} \left(\|u(t, \cdot; |\cdot|^{-\lambda})\|_\rho^2 \right) \leq 2(1 + Ct^{-\lambda}) \|\rho\|_{L^1(\mathbb{R}^d)} < \infty, \quad C = 2^{-\lambda} \left(\frac{\Gamma((d-\lambda)/2)}{\Gamma(d/2)} \right)^2,$$

Non-permissible $L_\rho^2(\mathbb{R}^d)$ -valued initial conditions:

(1) $\zeta(x) = |x|^\alpha$. When $\rho(x) = \exp(-|x|)$, then we can show that

$$C_1(1 + t^\alpha) \leq \mathcal{G}_\rho(t; |\zeta|) \leq C_2(1 + t^\alpha).$$

Comments about the initial conditions

- Previous theory was only able to consider $\zeta \in H = L^2_\rho(\mathbb{R}^d)$ initial conditions as candidates for the following limit:

$$\lim_{n \rightarrow \infty} \frac{1}{t_n} \int_0^{t_n} \mathcal{L}(u(s, \cdot; \zeta)) ds.$$

- Now $\mu \notin H$ is now allowed
- The heat operator smooths
- $u(t, \cdot; \mu) \in H$ for $t > 0$
- For $t > t_0 > 0$,

$$\mathcal{L}u(t, \cdot; \mu) = \mathcal{L}u(t - t_0, \cdot; u(t_0, \cdot; \mu))$$

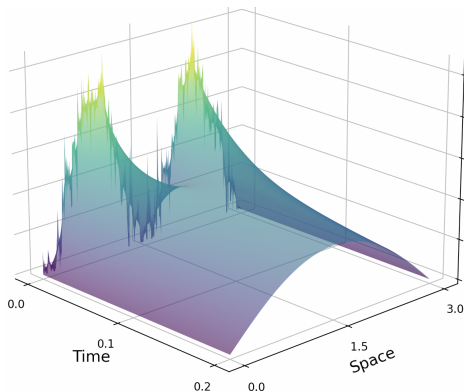


Figure: $u_t - u_{xx} = 0$ with a bumpy initial condition

Thank You!

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